Dose Finding, Efficacy Evaluation, and Why Programs Fail Late

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The two ends are already in the literature

Phase III comes in weaker than planned. 82 of 143 primary or coprimary survival endpoints, across 111 phase III oncology trials, had observed hazard ratios weaker than those assumed in the power calculations[1].

Dose questions outlive approval. 24 dose-optimization postmarketing requirements were issued for 21 of 138 newly approved oncology drugs. Among requirements that had resolved, the median time from issuance to fulfillment or release was 6 years[2].

Our work identifies one mechanism by which early decisions can create problems that surface later.

The thesis

We have been treating dose finding, phase II efficacy evaluation, and confirmation as three separate statistical problems.

Our results identify a **mechanism** by which optimizing those stages separately produces problems that only become visible later.

Weaker-than-expected confirmatory efficacy. Late-stage failure. Dose questions that survive into the post-approval setting.

What I am not arguing

- **Not** that poor dose finding explains phase III failure.
- **Not** that every phase boundary is unnecessary.
- **Not** that simulation itself is inadequate.

What I am arguing is narrower. Conventional development fixes architecture and tuning **before** asking which decisions and which evidence the final claim actually requires.

How a drug-development program usually works

Choose a dose
Which dose should move forward?

→

Look for an efficacy signal
Is it promising enough to continue?

→

Confirm efficacy
Can we establish that it works?

→

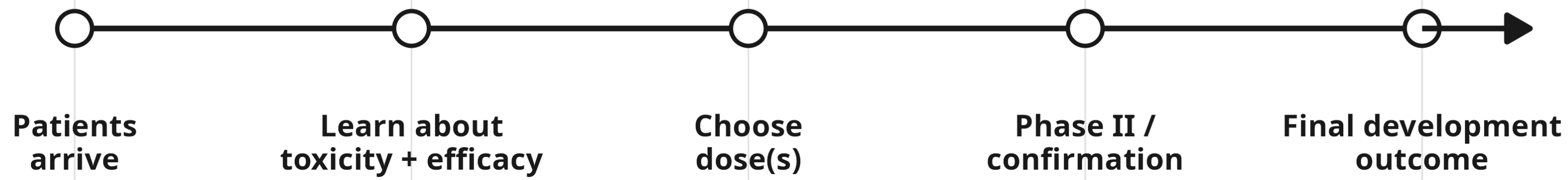
Approval
And the dose question does not always end here.

> Each transition fixes decisions and changes what evidence is still available later.

If a design's name does not determine its behaviour, what does?

Question 1 of 4

What did we actually simulate?



1. Start with established designs. BOIN, BOIN12, U-BOIN, each calibrated within its own family.

2. Simulate complete development programs. Not just which dose is selected, but what happens after selection.

3. Change one development decision at a time. When efficacy enters, how doses are ranked, which options remain, and what evidence can be reused.

Two questions, two levels of evaluation

Early-trial question		Development-program question
Did we select the correct dose?	↔	Did we ultimately confirm the correct dose?

Selection changes what evidence and options remain

First set of experiments: What features of a dose-finding design change the early decision?

Second set of experiments: What happens when that early decision is carried into different confirmation strategies?

The comparisons shown today use one application scenario in which efficacy increases with dose.

Terms used throughout

Clinical scenario. An assumed truth about the drug, the true toxicity and efficacy probability at every dose. The prespecified set spans efficacy increasing with dose, efficacy plateauing, a best dose adjacent to a sharply worse one, an inactive drug, and an overly toxic drug.

Correct dose. The dose with the best true benefit-risk tradeoff. Known to the simulation because the scenario specifies it. Never known to the design.

Calibrated. Tuning parameters and sample size searched against prespecified operating requirements. A candidate is then locked and independently re-verified on a fresh simulation stream. **The verdict is three-valued**, feasible, infeasible, or unresolved, and the next question is about why.

Design strategy. A design family, BOIN, BOIN12 or U-BOIN, together with its calibrated settings.

The phase II design used later in development. A standard two-stage design, stop early if too few responses, with a toxicity monitor.

Confirm. Declare efficacy at the end of that phase II design.

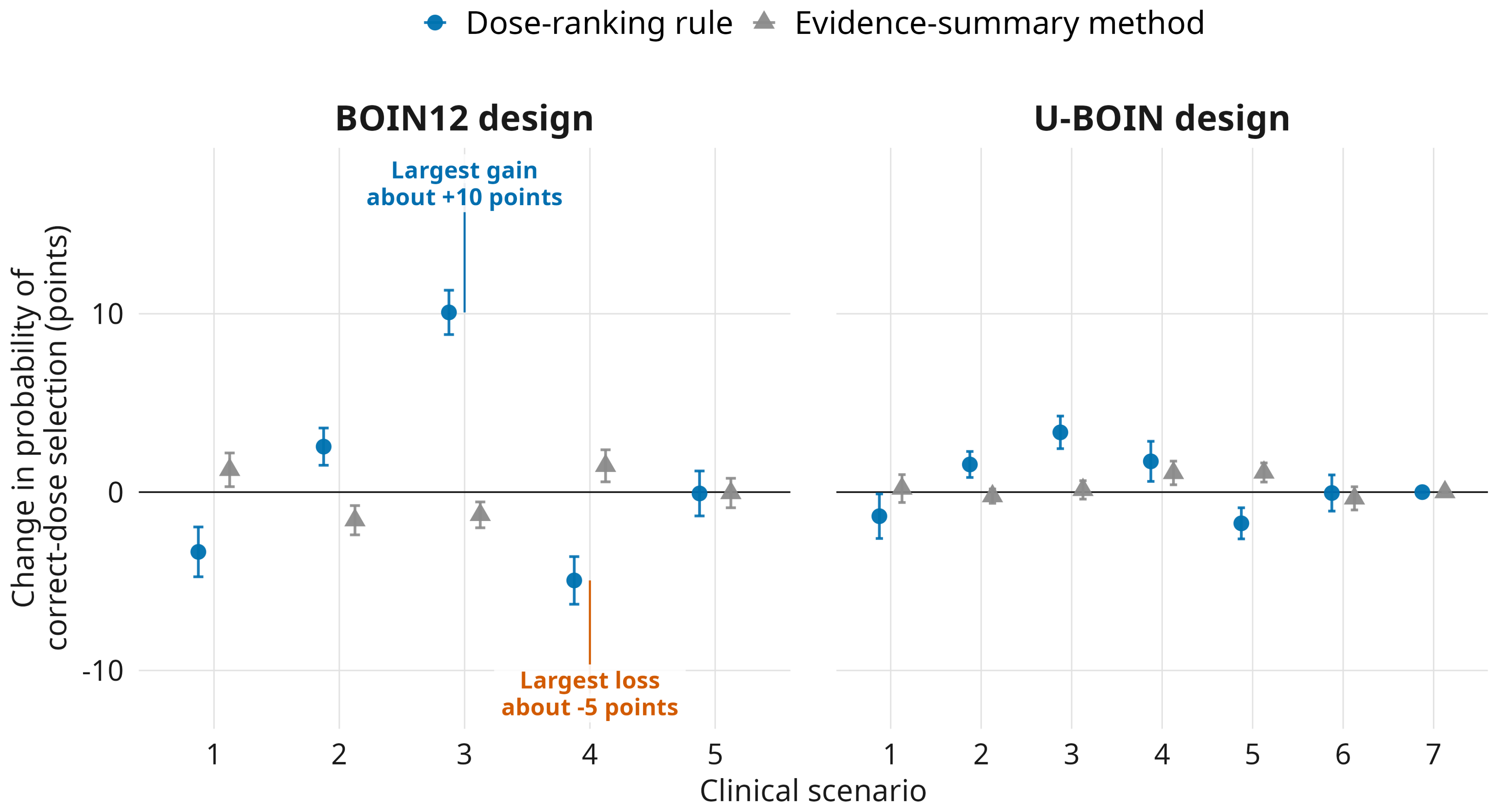
How to read the next result

What changes? Ranking rule versus evidence summary, one component at a time.

Why? If behavior changes when a component changes, the design label is too coarse.

Readout. Change in correct-dose selection on the same simulated patients.

What the design does matters more than what it is called



How to read the next result

What changes? Nothing. This one measures a property every variant shares.

Why? If the trial can permanently remove the best dose, no ranking rule can recover it.

Readout. How often the true best dose is eliminated before selection.

Some early mistakes cannot be repaired later

In this difficult scenario, the best dose is eliminated in about 1 in 10 trials.

Once it is gone, no later calibration can select it.

What Question 1 establishes

Performance is a property of components and context, not of a design's name.

Which raises the next question.

How would you establish that any particular implementation is good enough?

What would it take to establish that a design meets its requirements?

Question 2 of 4

The standard evaluation is a simulation table

For each of several scenarios it reports the probability of selecting each dose, the expected sample size, and the distribution of patients across doses.

That table is informative and it is what reviewers are accustomed to reading.

It reports quantities. It does not evaluate them against requirements.

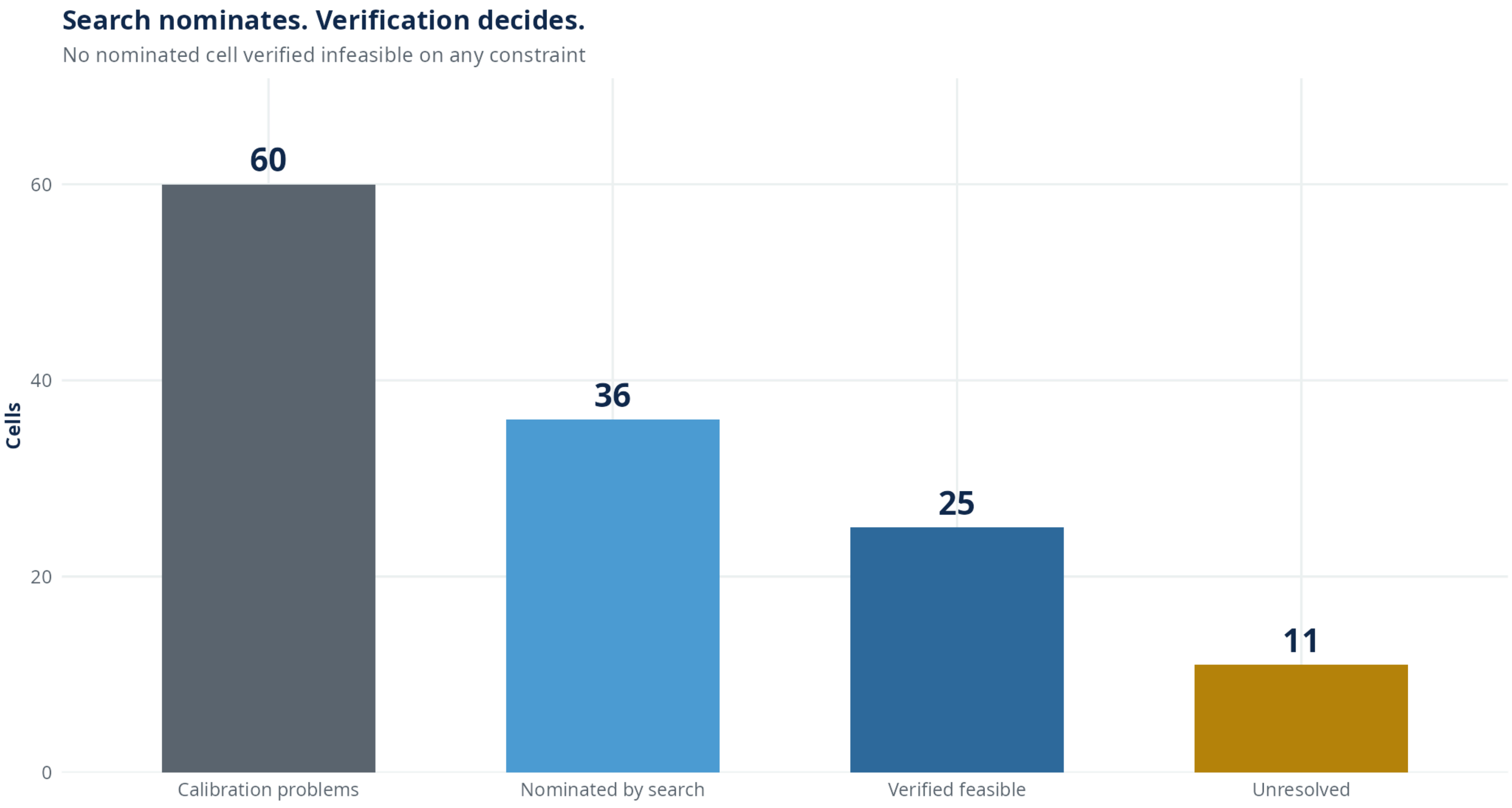
Nor does it report what was not tried.

What the table does not tell you

- **What value was required.** A reader sees the target selected in roughly half of trials. Required is not stated.
- **What else was available.** The tuning parameters define a space of alternatives. The table describes one point in it.
- **Whether the number is optimism.** When the same simulations tune and report the design, a favourable number carries optimism the table cannot show.

So the question becomes: what evidence would actually **establish** that a design meets its requirements?

Search nominates. Verification decides.



The distance is the framework working

The distance between what search nominates and what verification establishes is the framework operating as specified, not a loss.

Search selects on **point estimates**. Verification decides on an **interval**.

A nominated candidate whose remaining uncertainty still crosses the requirement is neither established feasible nor refuted on the evidence collected.

Reporting it as feasible would be exactly the selection-induced optimism the procedure exists to prevent.

Every verdict is three-valued

Verified feasible

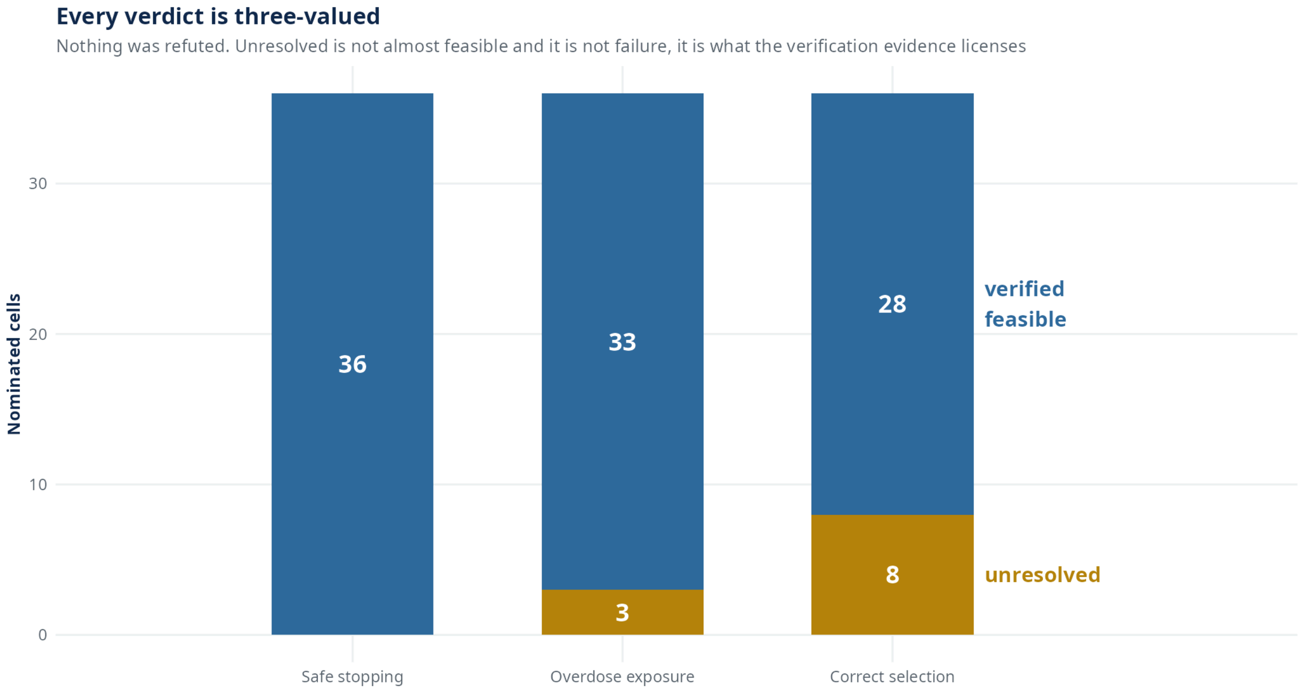
Evidence clears every prespecified requirement on the situations we chose in advance to test.

Unresolved

The interval still straddles at least one requirement. Not almost feasible. Not failure.

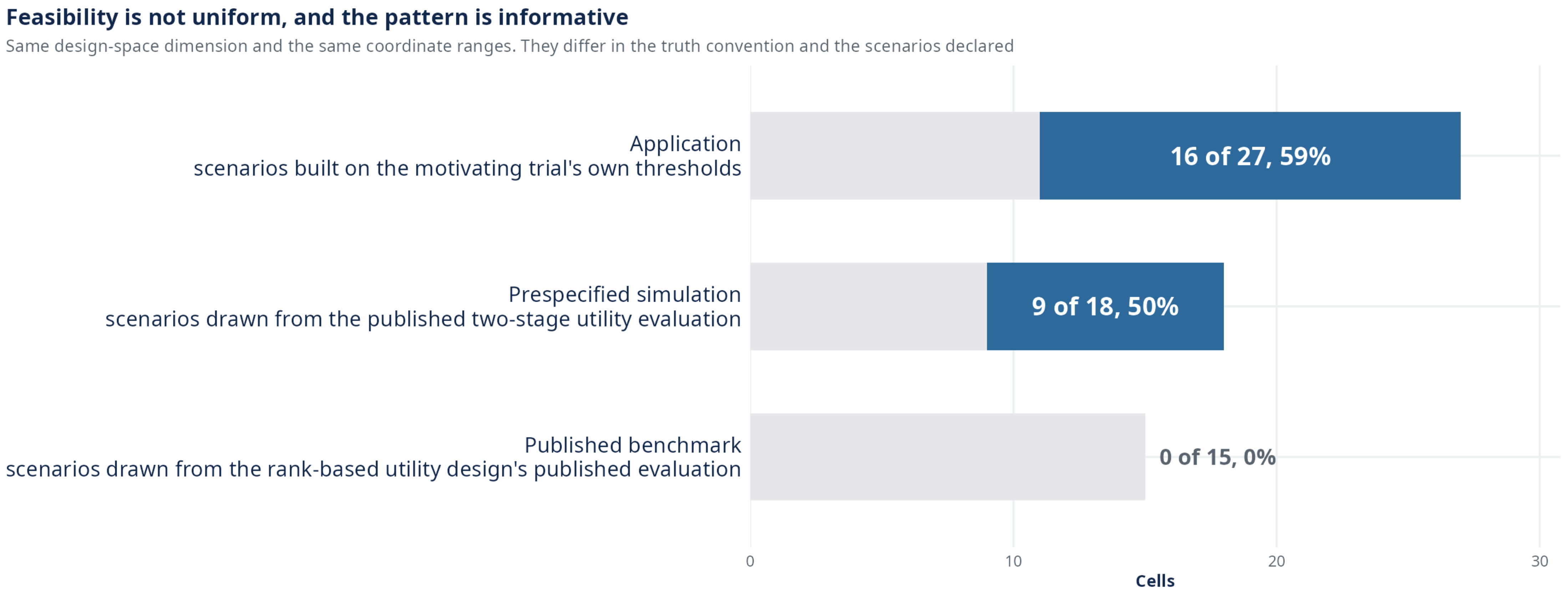
Verified infeasible

Evidence establishes that at least one requirement is missed.



Feasibility is not uniform, and the pattern is informative

Same design-space dimension and coordinate ranges. What changes is the truth convention and declared scenario set.



We offer **no mechanism** for the study that establishes none. The campaign as run does not isolate one.

A verification verdict is conditional on the situations we chose in advance to test

And that conditionality is a finding rather than a caveat.

Feasibility varied materially across sets of situations chosen in advance at a **fixed** search domain.

Locked designs re-scored on further scientifically plausible scenarios could cross constrained safety limits.

Verification on a prespecified scenario set does not establish robustness outside it. This is why the calibration specification is frozen and published with the design.

What a trial team receives

Design family	two-stage utility
Maximum sample size	27 patients, cohorts of 4
Starting dose	Level 3
Per-dose accrual cap	12 patients
Verification verdict	Verified feasible 5% error control across the 3 requirements checked together
Tuning vector	10 calibrated settings supplied with the protocol

This goes into the protocol, not just into the optimization code.

What a team does not receive is a guarantee

The verdict is a statement about **the scenarios the specification declared**, at the stated error level, for **the locked design**.

It is not a statement about the design family.

It is not a statement about scenarios outside the declared set.

It is not a statement about a wider range of designs than the one we searched.

Does the metric we optimize early match what the program needs?

Question 3 of 4

How to read the next result

What changes? Nothing about the designs. Only how we score them.

Why? Selecting the correct dose and ending development at the correct dose are different questions.

Readout. The two rankings, compared against each other.

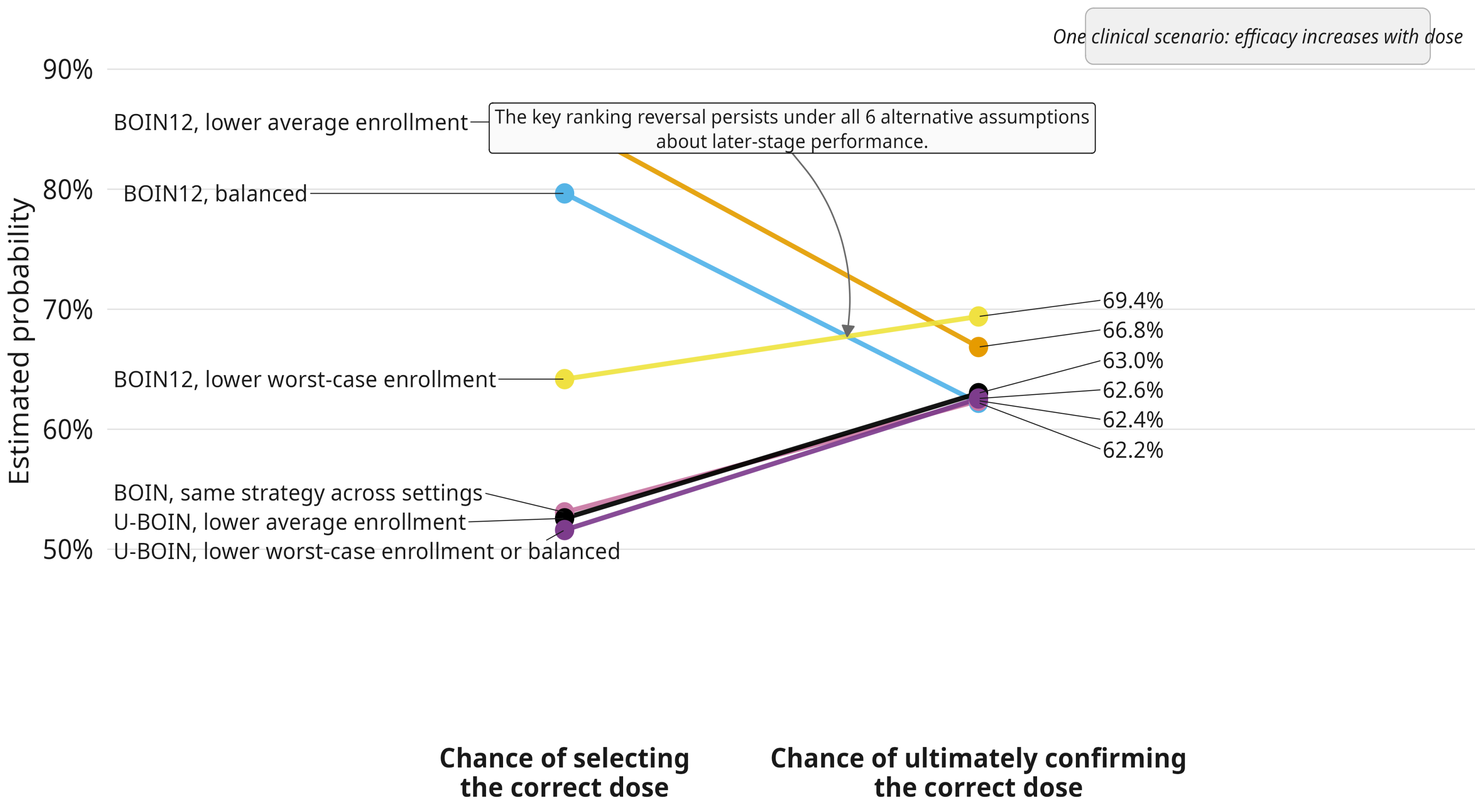
And the standard applies to us.

- Frozen downstream analysis set: **9 cells**.
- Corrected verification status: **7 verified feasible**, **2 unresolved**.
- We keep all 9, because changing the denominator after seeing the result would be changing the rules after seeing the results.

The locally best strategy need not be the globally best one

A strategy can score better at dose selection and worse after confirmation.

Correct-dose selection is an early objective used as a stand-in for ultimate program success.



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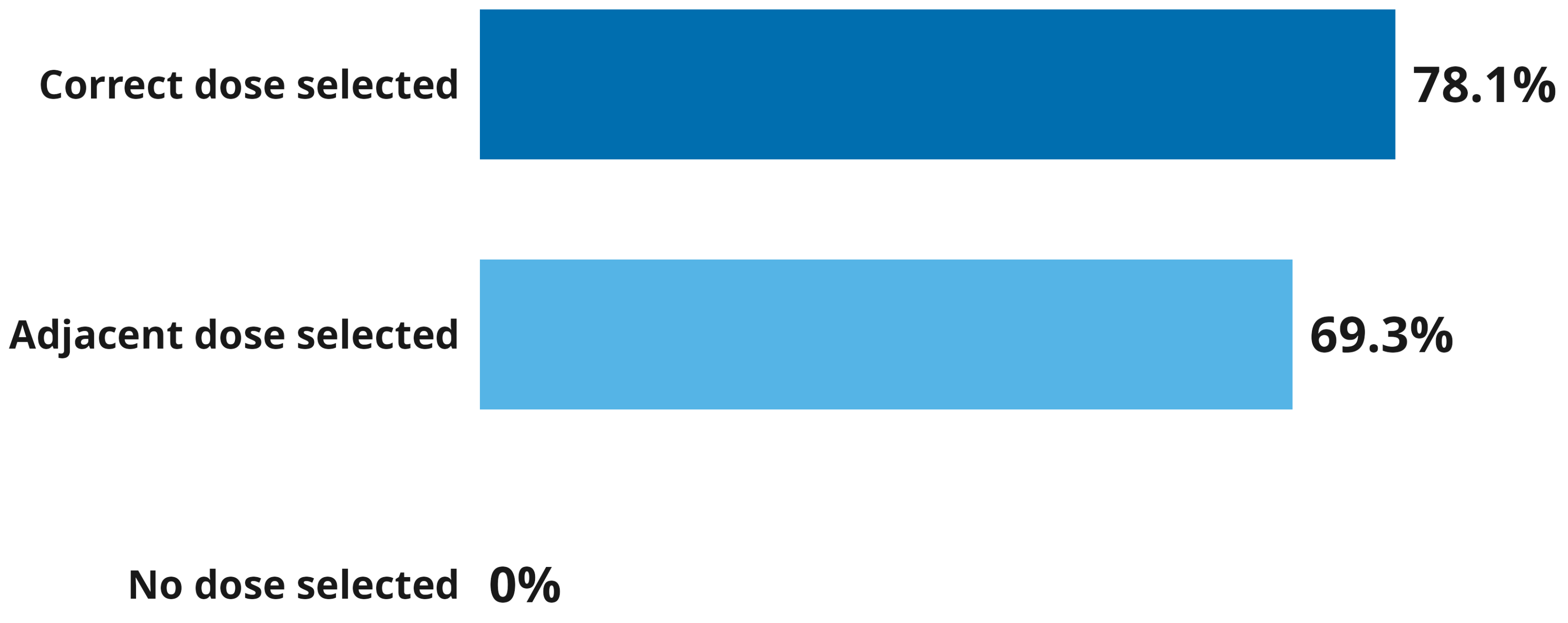
How to read the next result

What changes? How dose finding ends. The correct dose, the neighbouring dose, or no dose.

Why? Correct-selection probability scores the last two identically. Both are simply wrong.

Readout. What each ending is worth to the rest of the program.

Why correct-dose selection is not the same as program success



Correct-dose selection treats only the first outcome as success, even though the other outcomes have very different consequences later.

The second geometry bounds this result

That result is established on **one** truth. We repeated the question under a second response geometry, where efficacy plateaus instead of rising toward the target.

Only one design family produced enough eligible strategies to compare there. The families that carried the original inversion were not eligible on that geometry.

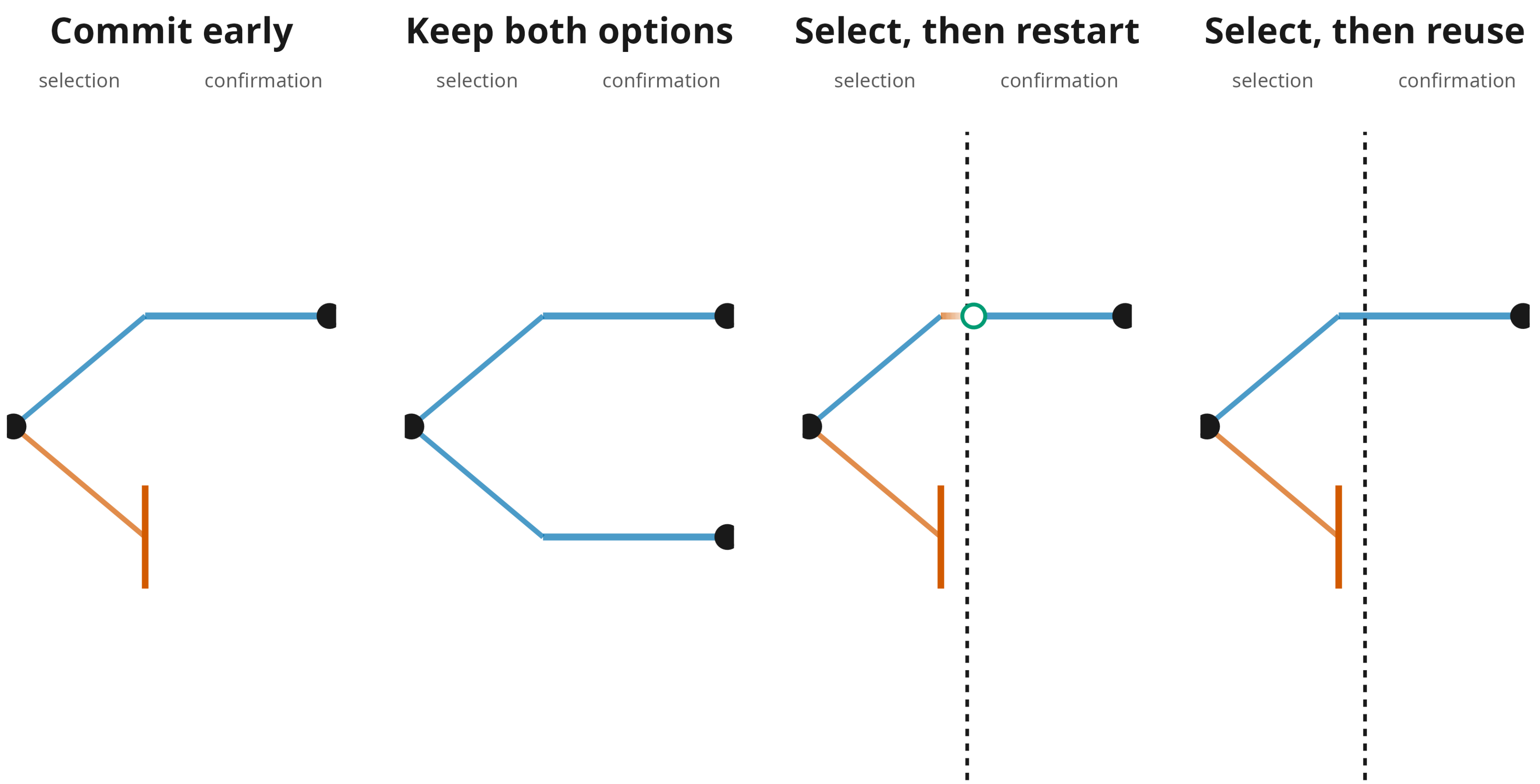
Within the eligible set, **the rank reversal did not recur**.

So the second geometry does not confirm the discordance result. It bounds it.

What does a phase boundary actually decide?

Question 4 of 4

A phase boundary determines what information survives



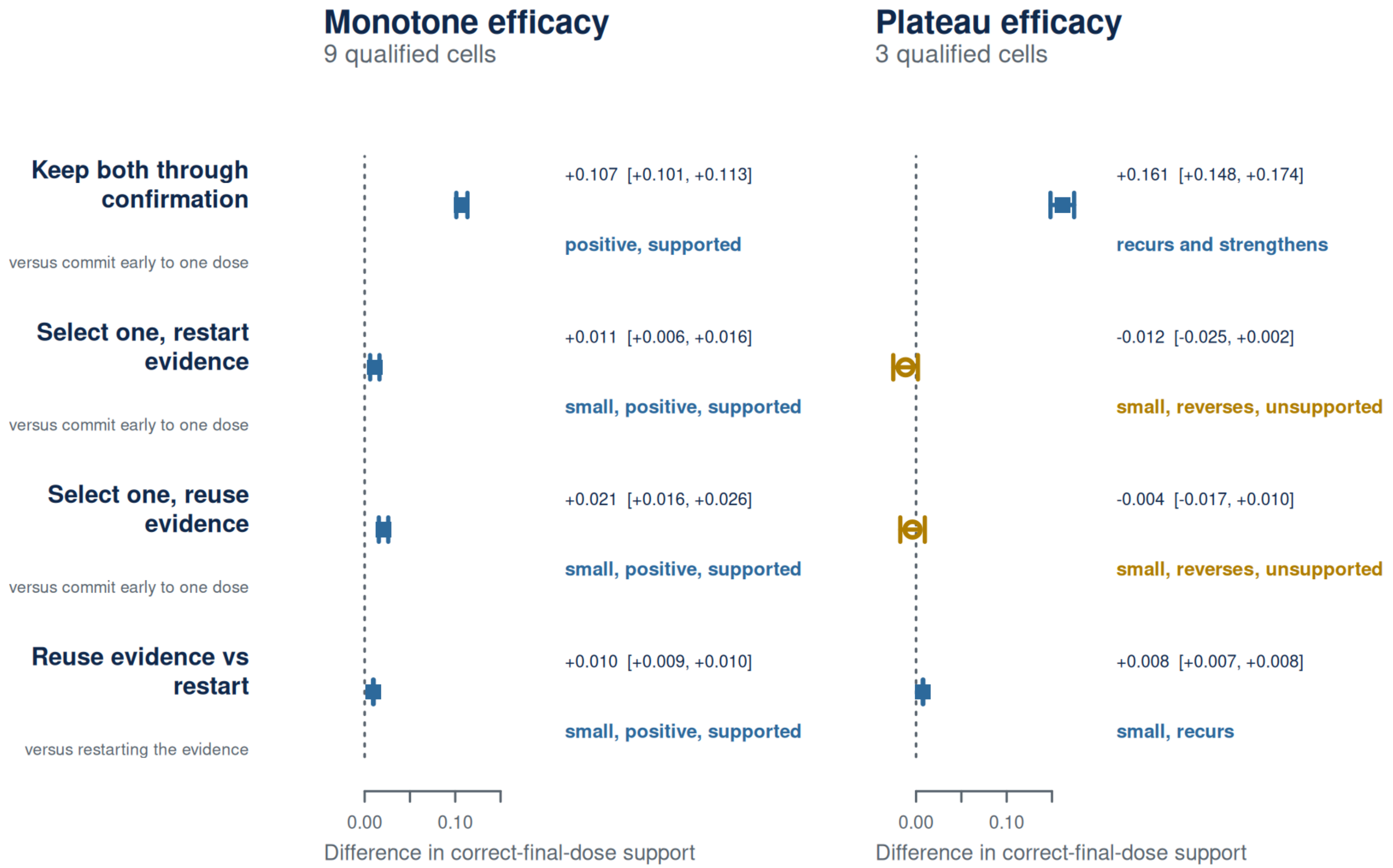
How to read the next result

What changes? Only what the program does after dose selection. The selections themselves are identical.

Why? FDA's Project Optimus encourages sponsors to compare more than one dose later in development.

Readout. Whether the correct final dose is supported after confirmation.

What changes across geometries



Filled marker, adjusted interval excludes zero. Open marker, it does not. "Small" means under one fifth of the largest effect shown.

What the geometry does to the mechanism

The mechanism used to resolve retained uncertainty has to match the shape of how efficacy changes across doses.

Using response data to choose between the two doses helps when efficacy separates the carried doses. When efficacy is flat across them, it no longer resolves the decision and can reverse direction.

So the transferable lesson is not “retain two doses”. It is retain uncertainty in a form the next evidence can actually resolve.

Some uncertainty can be bought back. Some information is gone.

Sotorasib shows what buying the answer later looks like

An approved lung-cancer drug whose 960 mg dose was compared against 240 mg only after approval. The original dose was not shown to be wrong. The dose question simply stayed open.

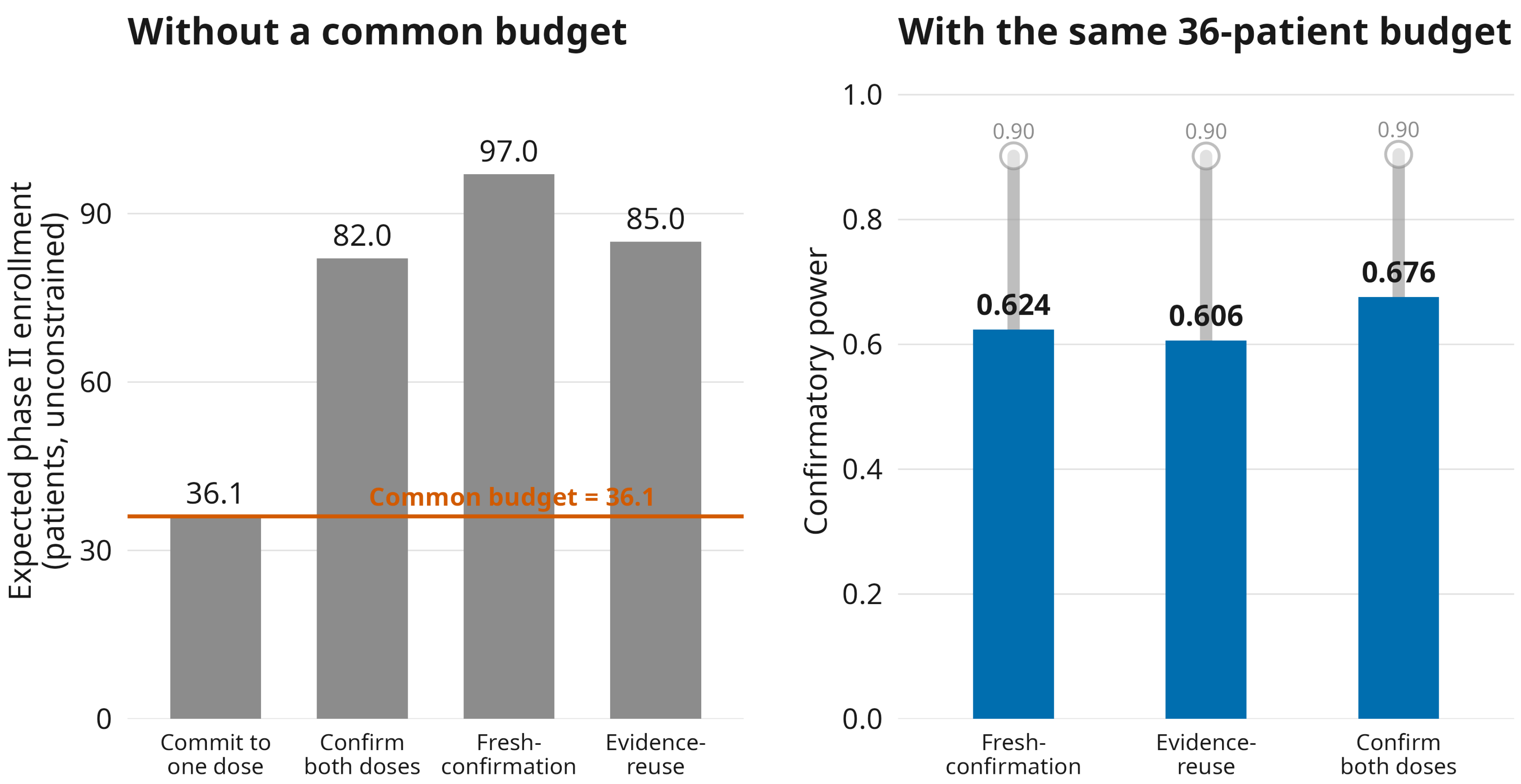
How to read the next result

What changes? Every strategy now gets the same patient budget.

Why? The obvious objection is that keeping two doses only helps because it spends more patients.

Readout. What each strategy can still achieve on that budget.

Preserving dose options has a cost



Preserving options has a cost. The question is where to pay it.

The systems hypothesis

Clinical development has been optimized **locally** while its errors propagate **globally**.

We have methods for optimizing individual trials, and even linked stages. We have not calibrated the full sequence around what early decisions leave later development able, and required, to prove.

A phase III analysis can have perfectly valid false-positive control while the development program has already failed scientifically.

Failed scientifically means the remaining program can no longer generate the evidence the final claim needs.

Where this leaves us

We have been treating dose finding, phase II efficacy evaluation, and confirmation as three separate statistical problems.

They are not separate. A dose-finding design's output is a dose **and the evidence and options it leaves behind**, and that state determines what the rest of development is able, and required, to prove.

Calibrate the design. Decompose what drives it. Then follow those components through the development program.

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